Data Preparation and Workbook Development

Implementation procedure

Project: Global Fuel Prices and Taxation Trends: A Comprehensive Analysis

Study period: 2020–2026 **Primary tool:** Tableau Public **Data source:** supplied global fuel-price CSV dataset

1. Source preservation

Keep an unchanged copy of the original CSV. Record its filename and the date it was received or last refreshed.

2. Initial inspection

- Confirm the number of rows and columns using the actual file.
- Inspect the date range and ensure dates are parsed correctly.
- Check whether numeric fields imported as numbers rather than text.
- Review nulls, duplicate records and unexpected values.
- Check category spelling and consistency for country, region and subsidy fields.

3. Field use

The date field supports time-series views; country and region support geographic comparison; income_level and subsidy_level can be used as contextual dimensions; petrol_usd_liter, diesel_usd_liter and lpg_usd_liter are fuel-price measures; brent_crude_usd and tax_percentage provide additional comparison measures.

4. Workbook development

Create one worksheet per analytical question. Use clear titles, consistent units, sensible number formatting, appropriate chart types and tooltips that identify the selected category and value.

5. Dashboard assembly

Place summary measures first, then detailed trend and comparison views. Keep filters discoverable and avoid making a viewer scroll unnecessarily.

6. Story development

Use a logical progression: introduce the question, describe the overall trend, compare regions or countries, then state the limitations and takeaway.

7. Recordkeeping

Record calculations, filters, exclusions and data-cleaning decisions so that another person can understand how each reported value was produced.

Prepared as a project-documentation draft. Replace illustrative wording with verified results, actual screenshots, and the student's own decisions before submission.